

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's 8 macro-domain / 41-subject taxonomy [Q42] aligns partially with central-exam priority subjects: Social Sciences and Law & Governance map to History/Polity, while Arts & Humanities covers cultural GK. However, the taxonomy does not separately enumerate Mathematics/Reasoning or Current Affairs as named categories — both heavily weighted in UPSC CSAT and SSC CGL Tier 1 [WEB-9, WEB-10, WEB-15]. Dataset analysis confirms reasoning and quantitative items exist (in the English partition: D8, D9), but the Hindi sample shows a heavy skew toward translated Engineering & Tech items (DC motor, half-wave rectifier, transformer, AM bandwidth, Fortran 77) that are peripheral to UPSC GS / SSC GA syllabi. State-level and central-level items are pooled without metadata distinguishing them [Q27], making it impossible to filter for central-exam scope [D24, D27].

ID	Evidence
Q27	"Our benchmark includes questions from over 40 different types of exams conducted both at the national and state levels over recent years." (p.3)
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
D8	Fill in the gap below with suitable word. Fruit : Reap:: Flower : ____... Bloom — Logical Reasoning/analogy question in English — confirms presence of reasoning subject
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition
D24	Which one of the following statements about Legislative Committee system of Maharashtra is not correct? ... Estimate Committee: All members of this committee are from Assembly only. — State-level (Maharashtra) governance question in English partition — not central exam scope
D27	मध्यप्रदेशके नई बन्दरगाह पर कौनसा नदी बहती है? ... मारवा — State-level GK (Madhya Pradesh industrial geography) in Telugu — regional, not central exam
WEB-9	Current Affairs is heavily and variably weighted in UPSC GS Paper 1, raising staleness concerns for older exam scrapes (https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage)
WEB-10	SSC CGL Tier 1 covers Reasoning, General Awareness, Quantitative Aptitude, English Comprehension (https://www.pw.live/ssc/exams/ssc-cgl-syllabus)
WEB-15	UPSC CSAT Paper 2 covers Quantitative Aptitude and Logical Reasoning (qualifying) (https://vajiramandravi.com/upsc-exam/csatsyllabus/)

Strengths

- Explicit India-first design covering Science, Social Sciences, Humanities, Arts, Business Studies, Law [Q23, Q74] aligns with multi-domain GK breadth required for UPSC/SSC.
- Hierarchical 8-domain × 41-subject structure [Q42] supports per-subject performance analysis relevant to syllabus mapping.
- Reasoning and quantitative aptitude content is present in MILU's taxonomy, confirmed empirically in the English partition [DATASET-D8, D9].

ID	Evidence
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Q23	“This benchmark covers many domains, including Science, Social Sciences, Humanities, Arts, Business Studies, and Law, among others.” (p.3)
Q42	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
Q74	“In this paper, we introduced MILU—Multilingual Indic Language Understanding Benchmark—a comprehensive benchmark specifically designed to evaluate LLMs across 11 Indic languages, spanning diverse domains and culturally relevant subjects.” (p.8)
D8	Fill in the gap below with suitable word. Fruit : Reap:: Flower : _____ Bloom — Logical Reasoning/analogy question in English — confirms presence of reasoning subject
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: For central-exam aspirants (UPSC/SSC/banking), priority categories are GK, Current Affairs, History, Polity, Mathematics/Reasoning, Hindi proficiency, Geography, Economics (elicitation Q1; [WEB-9, WEB-10, WEB-15]).

IO-2: Mathematics/Reasoning and Current Affairs are not named as MILU macro-domains [Q42]; they may be subsumed under other domains but lack explicit enumeration. Dataset findings confirm reasoning/quant items exist in the English partition [DATASET-D8, D9] but not whether the Hindi partition has comparable representation. State-level vs. central-exam labeling is absent [DATASET-D24, D27].

IO-3: MILU pools state-PSC and central-exam items [Q27, Q92] and the observed Hindi sample contains low-priority Engineering/Tech and niche CS items (Fortran 77, AM bandwidth) that are construct-irrelevant for UPSC/SSC GS [DATASET-D5, D6, D7, D16, D36].

IO-4: Two notable gaps: (a) absence of explicit Current Affairs and Mathematics/Reasoning macro-domains hampers syllabus-aligned scoring; (b) inability to separate central-exam from state-PSC items at the metadata level injects construct-irrelevant variance for this deployment.

ID	Evidence
Q1	“MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge.” (p.1)
Q27	“Our benchmark includes questions from over 40 different types of exams conducted both at the national and state levels over recent years.” (p.3)
Q42	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
Q92	“We collected our questions from over 40 exam types ranging from various National and state level civil service examinations to examinations conducted by various government and private organizations.” (p.17)
D5	जब एक डीसी सीरीज मोटर बना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering/Physics question in Hindi, `is_translated: True` — not in UPSC/SSC core syllabus
D6	हाफ वेव रेक्टिफायर का आउटपुट क्या होता है: ... प्लसेटिंग डीसी — Technical engineering question in Hindi, `is_translated: True` — domain not central to UPSC/SSC GS
D7	स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। ... शेल — Electrical engineering, translated — technical domain not prioritized for Hindi-medium UPSC prep

D8	Fill in the gap below with suitable word. Fruit : Reap:: Flower : ____... Bloom — Logical Reasoning/analogy question in English — confirms presence of reasoning subject
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition
D16	फॉरट्रान 77 के फकिस्ड फॉरमेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? ... एक जंप लेबल या फॉरमेट लेबल — Computer Science / Fortran programming question in Hindi — very niche for UPSC/SSC aspirants
D24	Which one of the following statements about Legislative Committee system of Maharashtra is not correct? ... Estimate Committee: All members of this committee are from Assembly only. — State-level (Maharashtra) governance question in English partition — not central exam scope
D27	మధ్యప్రదేశ్‌లోని బన్‌మూర్ హరికాండ అభివృద్ధి కేంద్రం ఏ జిల్లాలో ఉంది? ... మౌరసా — State-level GK (Madhya Pradesh industrial geography) in Telugu — regional, not central exam
D36	एएम की बैंडविड्थ _____ है। ... 1110 kHz — AM bandwidth engineering question — highly technical, 'is_translated: True'; unlikely in UPSC GS
WEB-9	Current Affairs is heavily and variably weighted in UPSC GS Paper 1, raising staleness concerns for older exam scrapes (https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage)
WEB-10	SSC CGL Tier 1 covers Reasoning, General Awareness, Quantitative Aptitude, English Comprehension (https://www.pw.live/ssc/exams/ssc-cgl-syllabus)
WEB-15	UPSC CSAT Paper 2 covers Quantitative Aptitude and Logical Reasoning (qualifying) (https://vajiramandravi.com/upsc-exam/csatsyllabus/)

Information Gaps

- Proportion of MILU Hindi items aligned to UPSC/SSC/banking vs. state PSC is not published.
- Whether Mathematics/Reasoning and Current Affairs are present in the Hindi partition at central-exam syllabus weight is not quantifiable from the small validation sample.

Requires Expert Verification

- A subject-matter expert in UPSC/SSC syllabus design should map MILU's 41 subjects to central-exam syllabus weights and quantify representation in the Hindi partition's test split.

Remediation

Gap 1: Mathematics/Reasoning and Current Affairs are not separately enumerated in MILU's 8 macro-domains; central-exam vs. state-PSC items are commingled without metadata.

Recommendation: Add metadata layer mapping each MILU item to (a) UPSC/SSC/banking syllabus category (including explicit Math/Reasoning, Current Affairs), (b) central-exam vs. state-PSC scope. This enables filtered evaluation aligned to syllabus weights [WEB-9, WEB-10, WEB-15].

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